

KUSH PATEL

Pittsburgh, PA

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🌐 [patelkush1508](https://patelkush1508.github.io)

🔗 [kushp15](https://github.com/kushp15)

Education

Carnegie Mellon University - GPA: 3.73/4.0

Pittsburgh, PA

Masters of Science in Artificial Intelligence & Information Security (Advanced Study)

May 2027

CourseWork: Machine Learning, Deep Learning, Advanced NLP, ML with Large Datasets, Trustworthy AI, Cloud Security, Distributed Systems, Secure Coding, Information Security, Advanced Data Networks

Rutgers University - New Brunswick - GPA: 3.67/4.0

Bachelor of Science in Computer Science & Minor in Business Administration

Sept 2022 – Aug 2024

CourseWork: Design & Analysis of Computer Algorithms, Artificial Intelligence, Deep Learning, Operating Systems & Design, Systems Programming, Computer Security

Technical Skills

Languages: Python, Go, C/C++, React, TypeScript

Technologies: PyTorch, TensorFlow/Keras, Langchain, Spark, JAX, OpenCV, Polars, FastAPI, Flask, Yolo

Others: Postgres, BigQuery, VectorDB, Git, Docker, AWS, GCP, Ubuntu, VSCode

Experience

Advantest America, Inc

South Plainfield, NJ

Software Engineer

Jun 2024 – Present

- Designed and optimized a low-latency computer vision and **OCR** pipeline, improving model accuracy by 13% and achieving sub-2 ms **ONNX** inference on live 4K video streams
- Built scalable ETL pipelines with **Apache Spark** and **dbt**, processing 500M+ records to power downstream ML and analytics workloads
- Integrated robotic **APIs** with legacy measurement software through a TCP/IP communication bridge, increasing automated inspection throughput by 90%
- Architected and deployed a **scalable** full-stack dashboard platform with **FastAPI**, caching, and optimized data processing for real-time applications

Technology Intern

Aug 2022 – May 2024

- Strengthened identity security for 500+ users through Azure AD IAM, MFA enforcement, and conditional access policies
- Automated audits and backup verification across 1000+ endpoints, streamlining compliance reporting and asset tracking

Projects

Multi-Layer Verification Prompt Injection Defense Pipeline | *Ollama, API, GCP*

[Link](#)

- Designed a **3-layer inference-time** defense pipeline (Structural Delimiters, Router LLM Firewall, and LLM-as-a-Judge) for LLM agents, cutting Direct Harm Indirect Prompt Injections on InjecAgent from 75.0% to 4.5% ASR
- Implemented a deterministic symbolic verifier and TF-IDF RAG threat-intelligence module, intercepting 97.7% of residual LLM-judge failure cases and reducing projected attack success to 0.4%

When to Stop Thinking: Adaptive Chain-of-Thought Reasoning in LLMs | *vLLM*

[Link](#)

- Developed a lightweight MLP probe over 4,096-dimensional LLM hidden states to predict answer readiness at each reasoning step, enabling adaptive chain-of-thought early exiting with ~0.80 ROC-AUC
- Built distributed inference and evaluation pipelines for long chain-of-thought workloads, achieving up to 98.1% simulated token savings across four reasoning benchmarks

GOD-100: Gap & Product Count Estimation in Dense Retail Environments | *DETR, Yolo, Pytorch*

[Link](#)

- Engineered SKU-110K-S (algorithmic gap generation via UniK3D depth) and manually annotated GOD-100 (10k+ objects, 1.3k+ gaps with physical depth measurements) to benchmark retail shelf void estimation
- Designed depth-fusion architectures (Faster-RCNN + UniK3D, FCOS + DA2) with custom regression heads to jointly detect shelf gaps and predict missing product counts, achieving 14.5 RMSE

CMU & Pittsburgh QA RAG System | *Search Engines, Crawl, Selenium*

[Link](#)

- Developed an end-to-end RAG system over scraped CMU and Pittsburgh cultural data, applying LoRA fine-tuning to all-MiniLM-L6-v2 dense embeddings and ms-marco-MiniLM-L-6-v2 cross-encoders for re-ranking

Lang IT | *Language Translation, GCP, VectorDB*

[Link](#)

- Integrated real-time overlay translated text on live screen sharing along with **RAG** for IT suport from live screen